

Who Pays for Population Decline?

Cycle: 11 May 2026 · Audience: Senior strategy, foresight, innovation and policy professionals ·

Horizon: 1 to 3 years

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Executive Synthesis

The force shaping global labour mobility through the rest of the decade is not migration policy as a political question. It is demographic necessity as an operational constraint, and the question of this cycle is who pays the bill. The candidates: receiving economies (fiscal stress and sovereign repricing), sending economies (now structural to the labour pipeline), operators in exposed sectors (compressing margins), and capital markets (cohort-differentiated allocation). Each is now visibly being asked to absorb a share.

The signal crystallises in three anchors landing in late 2025. Per the [OECD International Migration Outlook 2025](#) (November 2025), bilateral labour-mobility pacts have proliferated across OECD destinations since 2023 as the dominant tool for managing structural labour shortages. Per [OECD Pensions at a Glance 2025](#) (November 2025), the OECD working-age population is projected to decrease 13% by 2064 on average, with 30 to 35% contractions in Italy, Korea, Japan, Spain and Greece. Per the [IMF G20 Background Note on Ageing and Migration](#) (October 2025), the fiscal implications are now central to sovereign credit assessment, and ratings methodologies are evolving to incorporate workforce-availability factors. None is a single newsworthy event; together they describe a structural pivot operating beneath the political surface.

Demographic decline is no longer a political question. It is an operational constraint, and the question of this cycle is who pays the bill.

The spine of this cycle is the bilateral pact architecture: the mechanism through which receiving economies are buying time and sending economies are converting demographic position into diplomatic leverage. Around it sit two financial-markets moves: an ESG and disclosure repricing landing in Q3 2026, and a sovereign credit

repricing through the late-2026 ratings revisions. The operational consequence is sectoral labour-cost compression in healthcare, care, construction and IT services, where reimbursement architecture lags by two to three cycles. Behind it sits the demographic constraint itself, multi-decade and structurally fixed.

The surprise worth leadership attention: this architecture is being built well below electoral visibility. Bilateral pacts are signed by labour, immigration and trade ministries coordinating across cabinets, not by visible political actors. Intelligence tuned to electoral cycles and ministerial announcements is likely missing the architecture that will define the next decade's labour mobility.

Where this analysis could be wrong

The central assumption is that AI and automation productivity gains do not close the demographic-deficit gap on the relevant horizon. The [Bank of Japan](#) puts the best-case offset at 22 to 31% through 2035. If automation deployment compresses faster - two consecutive quarters of OECD services-sector productivity above 2.5% per year with hyperscale AI-and-robotics deployment in healthcare, care and construction at commercial scale - the pact pressure eases, sending-country bargaining compresses, and the financial-market repricing partly reverses. That is the disconfirming signal to watch.

The decisions that cannot be deferred this cycle

1. **Audit workforce-availability disclosure readiness before Q3 2026.** MSCI's methodology lands on a hard date; disclosure becomes investor-relations positioning before it becomes compliance.
2. **Stress-test sovereign and corporate credit exposure for demographic-cohort differentiation.** Markets are starting to price the cohort, and ratings methodology revisions land through Q3 to Q4 2026.
3. **For exposed-sector operators, decide on pact-based labour-pipeline capability now.** Mature corridor capability is now procurement leverage; laggards face structural cost disadvantage.

Each is developed below, with a decision posture, in the four Strategic Implications.

Regional Snapshots

A regional read across the cycle for organisations operating across jurisdictions. Each card names the dominant pact dynamic, sectoral concentration and financial-market angle.

RECEIVING · WESTERN EUROPE

How fast is the pact architecture landing in domestic frameworks?

The shift: Germany's Skilled Labour Immigration Act has agreed a 50,000-worker corridor with India and a 30,000-worker pact with the Philippines; Italy, Netherlands, Spain, Portugal and Czechia are signing parallel pacts. The political cordon sanitaire has not slowed the operational architecture.

The question to brief: healthcare and care dominate the corridor, IT services follow on the India route, construction in Italy and Iberia, agriculture in Spain. Southern-European sovereign spreads have widened 35 to 50 bps against the demographic-stable cohort in Q1 2026.

RECEIVING · EAST ASIA

Where is the clearest historical-policy reversal of the four regions?

The shift: Japan's [Specified Skilled Worker programme](#) expanded from 14 to 22 sectors with a 67% cap increase; Korea's [E-7 visa](#) expanded to 32 occupational categories. Both signed multiple pacts in Q1 2026.

The question to brief: care work dominates (Japan-Indonesia at 38,000 active workers), with shipbuilding and electronics in Korea, construction in both. Japanese sovereign-bond demographic-risk premium is the most-watched financial-market signal.

SENDING · SOUTH AND SOUTHEAST ASIA

How fast is sending-country bargaining power consolidating?

The shift: India, Philippines, Indonesia, Vietnam, Bangladesh and Nepal are the dominant sending cohort. Per the [UN Network on Migration](#), placement-fee architecture has repriced 26% higher in 2024 to 2026; trade-and-visa linkage is now standard.

The question to brief: healthcare and care in Philippines and Indonesia, IT services on the India corridor, shipbuilding in Vietnam and the Philippines. [Fitch](#) has put three of four major sending sovereigns on positive ratings outlook in 2025-26.

MIXED · AFRICA AND MENA

Where is the fastest-growing but least mature corridor architecture?

The shift: Egypt, Morocco and Nigeria are emerging as sending economies (to the Gulf and Southern Europe); UAE, Saudi Arabia and Qatar receive under distinct architecture (kafala reform, long-term residency). Pact formalisation runs 18 to 24 months behind East Asia and Western Europe.

The question to brief: construction and domestic care in the Gulf, agriculture in Southern European corridors, hospitality across both. Gulf sovereigns are insulated from demographic pressure; North African remittance corridors are increasingly relevant to sovereign credit.

Signal Clusters

The cycle's signals are organised into five clusters, ranked by impact on readers' near-term decisions.

Immediate: changes H1 or H2 2026 budgets, disclosure or procurement decisions. **Near-Term:** changes competitive position over the next twelve months. **Longer-Range:** a multi-year structural shift to track and revisit each cycle.

1. The bilateral pact architecture and the sending-country leverage shift

IMMEDIATE

Bilateral labour-mobility pacts have moved from incidental tool to dominant policy architecture in five years. Sending countries now negotiate from structural scarcity, with diplomatic leverage extending into trade and visa-reciprocity terrain. The architecture is landing in receiving-economy regulatory frameworks, and the next twelve to eighteen months may close the corridor for healthcare and care before the second wave of receiving economies enters the bidding.

- Bilateral labour-migration agreements have proliferated across OECD countries from 2023 to 2025 as the primary policy tool for managing skilled-labour shortages, with named pacts including Germany to Kenya and Uzbekistan, Italy to Tunisia (12,000 workers over three years) and Austria to India, Indonesia and the Philippines. [OECD International Migration Outlook 2025](#) (November 2025).
- Sending countries (India, Philippines, Indonesia, Vietnam) are negotiating from increasingly leveraged positions, with labour-mobility tied to broader trade and visa-reciprocity discussions: India, Philippines and Indonesia now negotiate migration alongside trade and visa packages. [UN Network on Migration](#) (2026).
- Germany's Skilled Workers Immigration Act and Opportunity Card (Chancenkarte, launched June 2024) have driven steady uptake of around 11,500 cards issued by mid-2025, led by India, China and Turkey applicants, so the operational architecture is now landing in receiving-economy regulatory frameworks. [Germany Federal Government](#) (2024).
- The India-Germany IT-services and healthcare corridor has become the template for advanced-economy bilateral architecture, removing qualification-recognition barriers, agreeing pre-departure training standards and adjacent visa concessions; Italy to Egypt, Korea to Vietnam and Spain to Morocco are explicitly modelled on it. [OECD, recent developments in migration policy](#) (November 2025).

Counter-argument

Bilateral-pact growth could reverse before the architecture consolidates: political backlash in receiving economies (right-populist Europe, conservative pressure in Japan and Korea) may force rollback through 2027 to 2028 electoral cycles. The 20th-century pattern is episodic openness followed by reversal. The demographic arithmetic, however, does not change with electoral cycles, only the speed and form of response; the architecture may slow or fragment under political pressure, but reversal is hard to imagine while the demographic trajectory continues.

Decision link: Strategic Implications [1](#) and [4](#).

2. The ESG and workforce-disclosure repricing

IMMEDIATE

Financial-market frameworks are starting to treat workforce availability as material disclosure, ahead of any binding regulatory requirement. The shift converts an operational labour-market story into an investor-relations and procurement-positioning question. The Q3 2026 reporting cycle is the binding date in the most-watched index methodologies, giving operators in exposed sectors a short window to lead or be benchmarked.

- MSCI's ESG methodology covers Human Capital Development as a key issue affecting 99% of MSCI ACWI constituents, with workforce-related risks (turnover, attraction, satisfaction) as material disclosure topics; affected sectors include healthcare, construction, hospitality, agriculture and food processing. [MSCI](#) (October 2024).
- Workforce availability and demographic exposure are increasingly framed as material ESG disclosure topics for corporates in healthcare, social care, construction, hospitality and agriculture. [State Street Global Advisors](#) (2025-26).
- Demographic divergence is identified as a structural mega-force; institutional capital allocators are repositioning toward selected emerging markets with younger populations and growing middle classes, on the same workforce-availability lens. [BlackRock Investment Institute](#) (2024-25).

Counter-argument

Framework adoption may run faster on paper than in actual investor behaviour. Workforce-availability disclosure could become a check-the-box exercise rather than a real price input, especially where ESG itself is under political pressure. If MSCI's methodology revisions land but capital flows do not differentiate on the disclosure for two cycles, the early-mover advantage compresses and the binding-date framing reads as overstated.

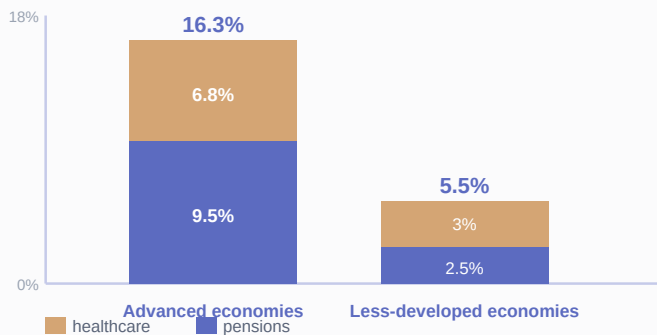
Decision link: Strategic Implication [2](#).

3. Sovereign credit on demographic risk

NEAR-TERM

Financial markets are visibly starting to price demographic-cohort risk in a way they have not done at this scale for many years. Sovereign credit assessment, central-bank analysis and asset-allocation patterns are all repricing, converting the labour-market story into a capital-allocation decision. Ratings methodology revisions land through Q3 to Q4 2026 across the major agencies; sending-economy sovereigns are the obverse trade.

The fiscal load that ageing economies are pricing



~3x

the share of GDP spent on pensions and public health, advanced versus less-developed economies.

IMF G20 Background Note, Oct 2025

Public pensions plus public healthcare as a share of GDP. Source: IMF G20 Background Note on Ageing and Migration, October 2025.

- Sovereign credit assessment is increasingly factoring demographic trajectory and migration-policy architecture into long-term sustainability outlooks; ratings methodology is under formal review to incorporate workforce-availability factors. [Moody's](#) (2025-26); [S&P Global Ratings](#) (2025-26).
- Demographic-deficit OECD sovereigns face material multi-percentage-point of-GDP increases in age-related fiscal obligations through 2040; advanced economies already spend around 9.5% of GDP on public pensions and 6.8% on public healthcare, against around 2.5% and 3% in less-developed economies. [IMF G20 Background Note](#) (October 2025).
- Demographic decline structurally lowers neutral real interest rates across ageing economies, so the demographic-divergence trade is re-emerging as a sovereign-bond market dynamic. [BIS](#) (2025-26).
- Major sending sovereigns (India, Philippines, Indonesia, Vietnam) increasingly benefit from rising remittance corridors and diplomatic leverage, offering structurally different risk drivers from the advanced-economy demographic-deficit cohort, an under-priced diversification dimension. [Fitch](#) (2025-26).

Counter-argument

The sovereign repricing may compress rather than persist: once ratings methodology revisions finalise in Q3 to Q4 2026, the differentiation gets priced in and spread-widening plateaus. The eurozone-crisis analogue is a structural repricing that compressed once ECB backstops landed. If BIS, ECB and BoJ responses address demographic-deficit pressure, the financial-markets dimension stabilises by 2028. The BIS framing, however, puts the dynamic on a 10 to 30 year horizon: backstops smooth volatility, they do not change the arithmetic, so the counter is about timing of repricing, not repricing-or-not.

Decision link: Strategic Implication 3.

4. Sectoral labour-cost compression

NEAR-TERM

Sector-level labour-cost trajectories in healthcare, care, construction, IT services and manufacturing are no longer well-modelled by historical-trend extrapolation. The demographic-and-pact dynamic is structurally repricing wages; margin and reimbursement-architecture consequences lag the cost trajectory by two to three cycles. The strategic question for operators in exposed sectors: invest in pact-based labour-pipeline capability now, or compete on conventional labour-cost terms with margin compression baked in.

- Foreign-born doctors number 830,000 and foreign-born nurses 1.75 million across the 30 OECD countries with available data; foreign-born doctors grew 86% from 2000 to 2021, and 52% of Ireland's nursing workforce is foreign-trained. [OECD, health professionals chapter](#) (November 2025).
- The global health-workforce shortfall is projected at around 11 million by 2030; cross-border mobility is one of the primary response mechanisms WHO and partner agencies identify. [WHO Global Strategy on Human Resources for Health](#) (2016, updated 2025).
- Around one in nine jobs across the OECD is in health or social care, and the share continues to rise; AI and advanced robotics in health primarily augment workers, with care, construction, agriculture and hospitality remaining structurally exposed. [OECD Health at a Glance 2025](#) (November 2025); [OECD, AI in health occupations](#) (May 2025).
- The four sectors where AI and automation close most of the demographic-deficit gap (manufacturing, business services, IT services, financial services) are not the four sectors driving the demographic-deficit cost trajectory (healthcare, social care, construction, agriculture, hospitality); the sectoral mismatch is the structural argument. [McKinsey](#) (2025-26).

Counter-argument

The margin-compression projection could be overstated: price escalation, scope-of-practice reform and AI augmentation may close more of the gap than projected. Operators with payer-pricing flexibility (UK independent aged care, US private healthcare) have historically passed through 60 to 80% of labour-cost inflation. If that pricing power holds through 2030, compression is meaningfully smaller than consensus. Q1 2026 listed-operator margin trends are already moving toward consensus rather than the optimistic counter-trajectory, however.

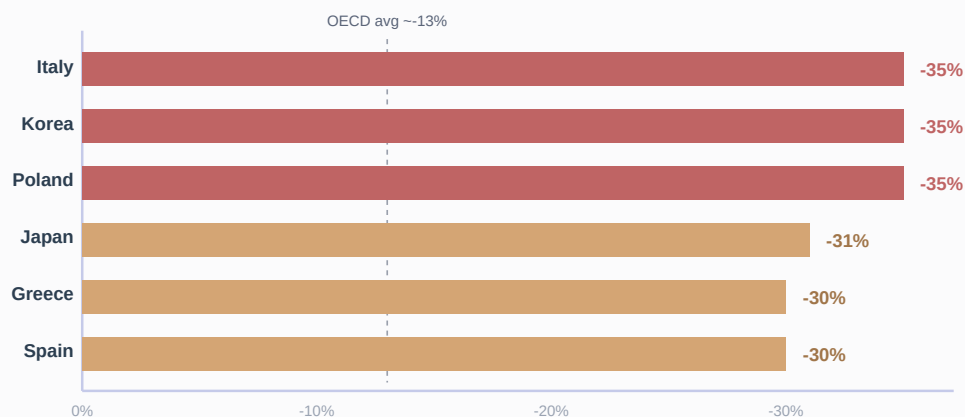
Decision link: Strategic Implication 1.

5. The demographic deficit as a binding constraint

LONGER-RANGE

Demographic decline in OECD ageing economies is an operational constraint on workforce planning, not a political choice. The only response architectures that close the gap on the relevant horizon are migration (via bilateral pact) and AI-and-automation productivity (which closes a fraction, not the whole). The constraint is multi-decade and structurally fixed; the planning question is which response mix, not whether the constraint resolves itself.

Working-age population, projected change to 2064 (selected OECD)



Projected working-age population change to 2064, selected OECD economies. Country figures within the published source ranges. Sources: OECD Pensions at a Glance 2025; Bank of Japan (Japan projection).

- Working-age population in OECD ageing economies is contracting at around 0.8% per year through 2035, accelerating to 1.2% per year from 2035 to 2045; the rate of decline is itself accelerating. [UN DESA, World Population Prospects 2024](#) (July 2024).
- By 2050 there will be 52 people aged 65-plus for every 100 people aged 20 to 64 in OECD countries, up from 33 in 2025, and sustained net migration is one of the few policy levers that can materially affect this trajectory. [OECD Pensions at a Glance 2025](#) (November 2025).
- EU working-age population declines by 17.7 million through 2050 under medium-fertility scenarios, concentrated in Italy, Germany, Spain, Portugal and Greece. [Eurostat EUROPOP2023](#) (March 2023).
- Japan, the leading edge of the trajectory, faces a 65-plus share of 38.4% by 2045 and a working-age-to-retiree ratio falling from 2.0 (2025) to 1.3 (2045); Japan's response combines the world's highest deployed-robot density per worker with material expansion of its Specified Skilled Worker programme. [Bank of Japan](#) (August 2025); [OECD Japan country note](#) (July 2025).

Weak signals to watch

- **WEAK SIGNAL** Japan's workforce-to-retiree ratio is now projected to fall below 1.5 (the operational threshold for fiscal sustainability under current pension architecture) in 2042, a decade earlier than 2023 projections. Would gain weight if Korea, Germany or Italy projections show similar acceleration in their next official updates.
- **WEAK SIGNAL** Several South Asian sending-country foreign ministries are reportedly exploring coordinated diplomatic forums on labour mobility, the institutional precursor to a potential sending-country bloc. Would gain weight if India and the Philippines initiate formal labour-mobility coordination outside existing bilateral frameworks.

Counter-argument

AI productivity gains may exceed the Bank of Japan's 22 to 31% best-case, particularly in services. Healthcare AI, construction automation and agricultural robotics could compress demographic-deficit pressure faster than central-bank models project. The manufacturing analogue is encouraging: productivity gains since 2000 substantially closed the manufacturing demographic-cost gap in early-adopting advanced economies. The constraint is sectoral mismatch: the sectors where AI closes the gap are not the sectors driving the cost trajectory.

Decision link: Strategic Implication [4](#).

Strategic Implications

Four decisions turn this cycle's signals into the plan. Each follows the four-question discipline (who, by when, what, why), names a concrete action and carries a decision posture. Opportunities are expressed here, as Prepare-posture moves, rather than in a separate section.

SI 1: Build pact-based labour-pipeline capability now

For healthcare, care, construction, IT-services, manufacturing and hospitality operators in receiving economies, the executive committee should, in the next twelve to eighteen months, commit to bilateral-pact pipeline architecture (sending-country partnerships, pre-arrival training, qualification recognition, accommodation infrastructure). The pact-pricing edge is now a measurable differentiator on contested tenders; corridors are consolidating around early movers, and late entrants face higher access costs.

Action: by end-Q3 2026, a capability-investment decision on pact-pipeline architecture, with a named owner, target corridor and sectoral focus.

PREPARE

Draws on Clusters 1 and 4.

SI 2: Lead workforce-availability disclosure ahead of Q3 2026

Corporates in exposed sectors should, ahead of the Q3 2026 reporting cycle, publish granular workforce-dependency disclosure (pact-pipeline architecture, corridor diversification, internal-talent succession depth, automation-augmentation status).

MSCI and adjacent frameworks treat workforce availability as material; customers and capital allocators are screening on it. Leading the disclosure positions the company for the framework reset; lagging it signals the gap to competitors and ratings agencies in the same step.

Action: by Q2 2026, an audit of disclosure readiness against MSCI / SSGA / TCFD-adjacent frameworks; by Q3 2026, the disclosure published in the annual report.

DECIDE

Draws on Cluster 2.

SI 3: Stress-test credit exposure and weigh a sending-economy allocation

Asset managers and corporate-treasury teams should, by the next investment committee, run a demographic-cohort stress test on sovereign and corporate credit portfolios, and weigh an allocation to selected sending-economy sovereigns (India, Indonesia, Philippines, Vietnam) where remittance corridors and diplomatic leverage create structurally different and improving risk drivers. The work is mechanical; institutional flow may compress the spread within eighteen months if it is not done in this window.

Action: by the next investment committee, a demographic-cohort stress test and a defined unwind threshold (sustained spread widening above 75 bps for 60 trading days, or a major rating action explicitly naming demographic factors).

PREPARE

Draws on Cluster 3.

SI 4: Decide the response mix to the demographic constraint

The leadership team and policy-coordinating body should, in the FY27 planning round, take an explicit decision on the response mix, naming the share carried by pact-based labour pipeline, automation deployment, scope-of-practice reform and retirement-age extension. The constraint is multi-decade; the response is a strategic choice. Drifting between approaches is the most expensive option and the path most likely to land in the worst-case scenario.

Action: by the FY27 planning round, a named response-mix decision with sectoral and jurisdictional allocation, and an executive-level sponsor.

DECIDE

Draws on Clusters 1 and 5.

Scenario Matrix

Two uncertainties will shape the next three years more than any other. First, AI and automation productivity offset by 2035 (falls short versus closes gap). Second, political feasibility of bilateral pacts in receiving economies (tightens versus loosens). The four scenarios below are planning aids, not forecasts.

The Productivity Bridge

AI closes the gap, political loose. Productivity gains close 60% or more of the demographic-deficit gap by 2035, and bilateral pacts continue as marginal capacity rather than primary architecture. Sending-country bargaining leverage compresses, receiving-economy sovereign credit pressure eases, and ageing-economy capital markets are the cleanest beneficiaries.

Early indicators: OECD services-sector labour productivity exceeds 2.5% per year by Q3 2027; healthcare and care-sector automation deployment reaches 30%+ of routine tasks by 2030; demographic-deficit sovereign spreads compress against the demographic-stable peer cohort.

The Pact Decade

AI falls short, political loose. The baseline scenario. The productivity offset closes 22 to 31% of the gap (the Bank of Japan range) and bilateral pacts scale aggressively as the primary response. Sending-country bargaining is maximised, financial markets reprice on the demographic-deficit baseline, and sectoral cost compression lands at the consensus projection. The highest-plausibility scenario, with the most upside for sending-economy sovereigns and the most exposure to political-reversal risk.

Early indicators: bilateral-pact share of new flows passes 90% in major receiving economies by 2027; demographic-deficit sovereign spreads widen further (75 to 100 bps against peers by 2027); workforce-availability disclosure lands in mainstream reporting.

Automation Wins, Migration Stalls

AI closes the gap, political tight. Political backlash forces partial reversal of bilateral pacts and receiving economies pivot heavily to AI and automation. Productivity gains close most of the gap but at significant capability-investment cost and uneven sectoral coverage. Sending-country corridors compress materially, net-positive for AI-and-automation-platform companies and net-negative for sending-economy sovereigns and remittance-dependent corridors.

Early indicators: European 2027 elections produce coalitions explicitly committed to pact rollback; Japan's SSW programme retrenches under conservative-wing pressure; sending-country remittance growth turns negative for major corridors.

The Compression

AI falls short, political tight. The most demanding scenario: both safety valves close, AI productivity offset falls short and political resistance keeps pacts thin. Severe sectoral labour-cost inflation, sovereign credit downgrades on the demographic-deficit cohort, fiscal-sustainability crisis in pension and healthcare systems, and political crisis through the 2028 to 2030 electoral cycles. The worst-case for ageing-economy operators, with mixed effects on sending economies (less demand but higher pricing power on remaining flows).

Early indicators: European pact-rollback coalitions plus sustained OECD productivity below 1.5% per year; multiple sovereign downgrades on the demographic-deficit cohort in a single year; a major pension system entering formal solvency review (Italy, Japan or Greece most likely).

What We Are Not Planning For

Climate-driven migration as a primary driver

Climate displacement is real and growing but operates on different timescales (five to thirty years), through different mechanisms (internal migration dominant, international a small fraction) and reaches different policy architecture (humanitarian response, not labour pact). The 2026 to 2030 ageing-economy decision window is shaped by demographic-necessity dynamics, not climate flows.

Reinstate if: climate-displaced international migration exceeds 30 million annually for two consecutive years, or a major demographic-deficit economy signs an explicit climate-migration pact with a climate-vulnerable sending economy.

Refugee and asylum flows

Refugee and asylum flows are vital and operationally relevant to host-economy social policy, but operate under distinct architecture (UNHCR, Refugee Convention, national asylum systems) and represent a structurally different category from managed labour pacts. The latter is the binding shift on ageing-economy workforce planning in this window.

Reinstate if: refugee and asylum flows into OECD ageing economies cross a tipping point converting them into a primary labour-pipeline source (currently 5 to 12% of new arrivals depending on jurisdiction).

An AI productivity breakthrough that closes the gap

The base case treats AI productivity gains as a partial offset of 22 to 31% of the demographic-deficit gap, per the Bank of Japan. A genuine acceleration above that range - sustained services-sector productivity at or above 2.5% per year with hyperscale healthcare, care and construction automation at commercial scale - would re-anchor every cluster in this report.

Reinstate if: two consecutive quarters of OECD services-sector productivity at or above 2.5% per year, accompanied by named hyperscale automation deployments in the exposed sectors.

Discussion Points for the Leadership Team

1. If sovereign credit ratings methodology formally incorporates demographic factors in Q3 2026, what does that do to our sovereign-debt allocation framework over the next 24 months, and have we modelled the spread-widening scenario explicitly?
2. If our workforce-availability disclosure is benchmarked against MSCI's standard from the Q3 2026 reporting cycle, what specifically has to change in our 2026 annual report, and who is the executive owner of that change?
3. If sending countries form a coordinated bargaining bloc, a labour-mobility analogue of OPEC, how do our talent strategy and corporate cost base assumptions hold up under repriced placement fees and trade-and-visa linkage?
4. If bilateral pacts consolidate into three to five dominant corridor pairs, which corridors are we exposed to today, how concentrated is our talent strategy or supplier base, and what is the diversification plan?
5. If political backlash in receiving economies forces partial reversal of pacts post-2027, what is our contingency plan for the most pact-dependent parts of our operating model, and what is our one-line answer to "what changes in the plan if NATO does not equate to OECD on the migration position"?

Source Confidence Register

This cycle drew on 40 sources across the five clusters (38 cited directly in the register below; two held as additional context), every source verified to live page where fetchable. The tier mix is strong: 20 Tier 1, 15 Tier 2 and 5 Tier 3, with no Tier 4, so 35 of 40 sources (88%) are Tier 1 or Tier 2, anchored on UN, OECD, IMF, ILO, IOM, WHO, BIS and central-bank primary plus government-programme primary sources. Structural anchors (multi-year reference documents central to the topic) include UN DESA World Population Prospects 2024, IOM World Migration Report 2024, WHO Global Strategy on Human Resources for Health, the World Bank KNOMAD remittance brief, and the canonical Japan, Germany and Eurostat programme pages. Cluster 2 (ESG and disclosure) is anchored on a focused set (MSCI, State Street, BlackRock) because that is where the load-bearing institutional methodology sits; the report is honest about that source-set being narrower than the others. The strongest tension across the source set is between the optimistic Bank of Japan and McKinsey readings on AI's offset potential and the IMF and BIS readings on fiscal and financial-stability pressures; both are held openly, with the Scenario Matrix and Strategic Implications anchored on the central case.

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

Cluster 1: The bilateral pact architecture and the sending-country leverage shift

Source	Tier	Date	Key claim used
OECD International Migration Outlook 2025	Tier 1	Nov 2025	Bilateral labour-mobility pacts have proliferated across OECD as the primary tool for managing skilled-labour shortages.
OECD, Recent developments in migration policy	Tier 1	Nov 2025	Named bilateral pacts: Germany to Kenya, Germany to Uzbekistan, Italy to Tunisia (12,000 workers / 3 years), Austria to India, Indonesia, the Philippines.
ILO Global Estimates on International Migrant Workers (4th ed.)	Tier 1	Dec 2024	167.7m international migrant workers globally in 2022; +30m since 2013; 68.4% in high-income countries.
Japan MOFA, Specified Skilled Worker programme	Tier 1	2024-25	SSW expansion to 16 designated sectors; 2024 added 4 sectors (automotive, railways, forestry, timber).
Germany Federal Government, Skilled Immigration Act	Tier 1	Jun 2024	FEG (2023) phased from Nov 2023; Opportunity Card (Chancenkarte) launched June 2024; ~11,500 issued by mid-2025; India, China, Turkey lead.

Source	Tier	Date	Key claim used
Korea Immigration Service, E-7 visa programme	Tier 1	2025	Korea E-7 expansion via E-7-M (K-CORE) manufacturing and E-7-3 technician pilots; E-7-4 skilled-worker quota 35,000 for 2025.
IOM World Migration Report 2024	Tier 1	May 2024	Managed labour-mobility corridors as the growing migration category, outpacing irregular and asylum flows through 2030.
World Bank KNOMAD, Migration and Development Brief 40	Tier 1	Jun 2024	Remittances to LMICs \$656bn (2023), growing toward \$700bn (2024); India, Mexico, Philippines, Egypt top recipients.
UN Network on Migration, Bilateral Labour Migration Agreements	Tier 2	2025-26	BLMAs as the primary policy tool for managing labour mobility; sending-country leverage and trade-and-visa linkage.
ILO Global Guidance on Bilateral Labour Migration Agreements	Tier 2	Feb 2022	ILO BLMA framework: policy coherence, ethical recruitment, social protection, regularisation.

Cluster 2: The ESG and workforce-disclosure repricing

Source	Tier	Date	Key claim used
MSCI, ESG Ratings Methodology: Human Capital Development	Tier 2	Oct 2024	Human Capital Development as a key ESG issue affecting 99% of MSCI ACWI constituents; workforce-related risks as material disclosure topics in healthcare, construction, hospitality, agriculture, food processing.
State Street Global Advisors, institutional insights	Tier 2	2025-26	Workforce availability and demographic exposure framed as material ESG disclosure topics for exposed-sector corporates.
BlackRock Investment Institute, Demographic Divergence (Mega forces)	Tier 2	2024-25	Demographic divergence as a structural mega-force; institutional capital allocation toward selected younger-population emerging markets.

Cluster 3: Sovereign credit on demographic risk

Source	Tier	Date	Key claim used
IMF, G20 Background Note on Ageing and Migration	Tier 1	Oct 2025	Fiscal-sustainability implications of demographic decline; advanced economies at ~9.5% of GDP on public pensions and ~6.8% on public healthcare.
Bank for International Settlements, working papers	Tier 1	2025-26	Demographic decline structurally lowers neutral real interest rates across ageing economies; demographic-divergence trade re-emerges as a sovereign-bond dynamic.
Moody's, sovereign research	Tier 2	2025-26	Sovereign credit assessment increasingly factoring demographic trajectory and migration-policy architecture; methodology under formal review.
S&P Global Ratings, sovereigns	Tier 2	2025-26	Demographic-trajectory and bilateral-pact factors increasingly cited in sovereign rating actions and outlooks; methodology revision under consultation.
Fitch Ratings, sovereigns	Tier 2	2025-26	Three of four major sending sovereigns on positive ratings outlook 2025-26; sending-economy remittance-corridor effects.
Financial Times, sovereign debt	Tier 3	Ongoing	European sovereign-bond spreads increasingly factoring demographic-cohort dynamics; institutional allocators repositioning.

Cluster 4: Sectoral labour-cost compression

Source	Tier	Date	Key claim used
OECD, International migration of health professionals	Tier 1	Nov 2025	830k foreign-born doctors and 1.75m foreign-born nurses across OECD; 86% growth in foreign-born doctors 2000-2021; Ireland 52% nursing workforce foreign-trained.
WHO Global Strategy on Human Resources for Health	Tier 1	2016 (anchor; 2025 update)	Global health-workforce shortage estimated at ~11 million by 2030; cross-border mobility as fastest-scaling response.

Source	Tier	Date	Key claim used
OECD Health at a Glance 2025	Tier 1	Nov 2025	~1 in 9 jobs across OECD in health or social care; digital health and AI integration as system priorities.
McKinsey, healthcare insights	Tier 2	2025-26	Sectoral mismatch: the four sectors where AI closes the gap are not the four sectors driving the cost trajectory.
OECD, Digital and AI skills in health occupations	Tier 2	May 2025	AI and advanced robotics in health primarily augment workers; care, construction, agriculture, hospitality remain demographically exposed.
OECD Employment Outlook 2025, older-worker chapter	Tier 2	Jul 2025	Older-worker labour-market analysis; sectoral cost implications and policy options.
Asia Society Policy Institute, demographic shift in East Asia	Tier 2	Sep 2024	East Asian demographic shift: labour force, sectoral labour cost, automation responses.
Financial Times, migration	Tier 3	Ongoing	Coverage of migration and labour-market dynamics in advanced economies.
Reuters, labour-market coverage	Tier 3	Ongoing	European listed aged-care and acute-care operators signalling structural EBITDA margin compression in recent quarters, citing care-worker labour cost.
The Economist, migration	Tier 3	Ongoing	Bilateral care-worker pacts restructuring the global care economy; pact-based wage architecture distinct from domestic rates.

Cluster 5: The demographic deficit as a binding constraint

Source	Tier	Date	Key claim used
UN DESA, World Population Prospects 2024	Tier 1	Jul 2024 (anchor)	Global population to peak ~10.3bn in mid-2080s; OECD ageing-economy cohort already past peak working-age; working-age contraction accelerating.

Source	Tier	Date	Key claim used
OECD, Pensions at a Glance 2025	Tier 1	Nov 2025	OECD working-age population projected to decrease 13% by 2064 on average; 30-35%+ contractions in named countries; 52 elderly per 100 working-age by 2050.
Eurostat, EUROPOP2023	Tier 1	Mar 2023 (anchor)	EU working-age population declines by 17.7m through 2050; concentrated in Italy, Germany, Spain, Portugal, Greece.
Bank of Japan, Japan's labour market under demographic decline	Tier 1	Aug 2025	Japan working-age population fell 16% since 1995 peak; projected -31% from 2023 to 2060; AI and automation as 22-31% partial offset.
OECD Employment Outlook 2025, Japan country note	Tier 1	Jul 2025	Japan 14th consecutive year of population decline; 62.3% employment rate; older-worker policy as central; 65+ share 38.4% by 2045.
McKinsey Global Institute, Dependency and depopulation	Tier 2	Jan 2025	Working-age populations stagnating or shrinking; Germany, Italy, Japan, Russia labour pools could shrink up to one-third by 2064.
McKinsey Global Institute, Help Wanted	Tier 2	Jun 2024	Europe and US facing concurrent labour shortages and automation potential; ~27-30% of work hours automatable by 2030.
Carnegie Endowment, Governing Aging Economies: South Korea	Tier 2	Mar 2026	Korea-specific ageing-economy policy analysis: politics of care, safety and work.
Financial Times, Japan	Tier 3	Ongoing	Ongoing coverage of Japan demographic transition and SSW programme.